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Design of a SCR system for power plant exhaust gas treatment

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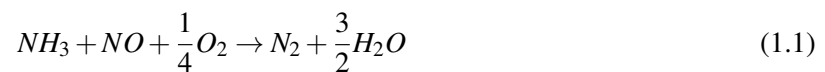
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1 Introduction

1.1 Description

The design of a SCR system for power plant exhaust gas treatment was carried out in this project.

The SCR system's main objective is to reduce the emissions of NO_x present in flue gases, due to their toxic nature. This is done through the insertion of NH_3 at the SCR inlet, which will then react inside the catalyst, together with NO_x , reducing it. In (Eq. 1.1) is present the chemical reaction that occurs inside the catalyst.



The objectives of this project were multiple, but they can be summarised as follows:

- Compute the catalytic reactor's dimensions, considering only the pressure drop, with some geometrical limitations.
- Choose a suitable turbulence model, discretization scheme, specify flow regimes in the different regions of the reactor.
- Choose the best position and number of ammonia injectors to optimize the reaction.
- Test the model for different operation conditions.

1.2 Assumptions

In order to reduce the complexity of the problem and the computational time needed to model the SCR, several assumptions and simplifications were made. To begin with, the evaporation of the spray inside the reactor was not considered, and at the same time the reactor's geometry was simplified. Instead of modelling the entire U-turn, only the final part was modelled, where the catalysts are present, as shown in (Fig. 1). In this way also the symmetry can be used to further simplify the computations.

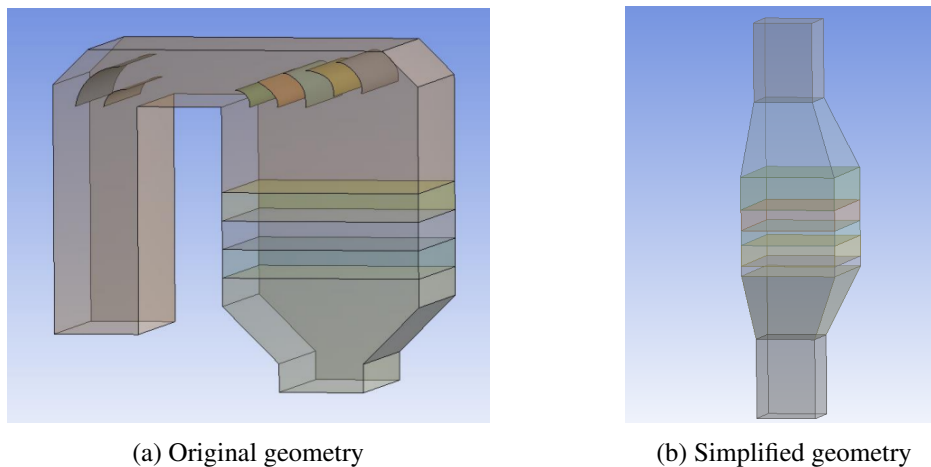


Figure 1: Geometry comparison

1.3 Constraints and data

Several limitations were imposed on the problem, such as:

- Maximum reactor pressure drop: $\Delta P = 600 \text{ Pa}$.
- Maximum ammonia slip: $\text{NH}_3 = 15 \text{ ppm}$.
- Maximum reactor height: $Z = 20 \text{ m}$.
- Inlet/outlet channels must be $6 \text{ m} \times 6 \text{ m}$ and at least 4 m long.
- Maximum number of ammonia injectors: 5.

Furthermore the catalyst was divided into modules, elements and channels, whose dimensions and structure were given. The number of modules that constitute the catalyst was used as design parameter, its value was computed in order to fulfill the requirement on the pressure drop, considering the case at 100% of production, in order to be conservative.

1.4 Operating conditions

The SCR has been tested at different operating conditions of the powerplant, in all of which it must always have a maximum pressure drop of $\Delta P = 600 \text{ Pa}$, and maximum ammonia slip of 15 ppm . The operating conditions are shown in (Tab. 1).

% of full production	25%	50%	75%	100%
Flow [Nm^3/h]	206000	356000	493000	683000
Temperature [$^\circ\text{C}$]	290	310	340	365
Pressure [Pa]	99650	99400	99000	98400
Mole fraction of NO	3.8×10^{-4}	4.2×10^{-4}	4.8×10^{-4}	4.9×10^{-4}
Mole fraction of H_2O	0.125	0.125	0.125	0.125
Mole fraction of O_2	0.0625	0.0625	0.0625	0.0625
Mole fraction of CO_2	0.0625	0.0625	0.0625	0.0625

Table 1: Powerplant operating conditions

2 Theory

2.1 Flow equations

In order to solve the flow, several equations need to be used:

- **Continuity equation**

A material balance over a stationary fluid element $\Delta x_1 \Delta x_2 \Delta x_3$ is written

$$\{\text{accumulation}\} = \{\text{transport in}\} - \{\text{transport out}\} \quad (2.1)$$

$$\begin{aligned} \Delta x_1 \Delta x_2 \Delta x_3 \frac{\partial \rho}{\partial t} = & \Delta x_2 \Delta x_3 [(\rho U_1)|_{x_1} - (\rho U_1)|_{x_1 + \Delta x_1}] \\ & + \Delta x_1 \Delta x_3 [(\rho U_2)|_{x_2} - (\rho U_2)|_{x_2 + \Delta x_2}] \\ & + \Delta x_1 \Delta x_2 [(\rho U_3)|_{x_3} - (\rho U_3)|_{x_3 + \Delta x_3}] \end{aligned} \quad (2.2)$$

This formulation corresponds to the finite-volume formulation of the continuity equation for a hexahedral mesh aligned with the coordinate axis. Dividing by $\Delta x_1 \Delta x_2 \Delta x_3$ and taking the limit $\Delta x \rightarrow 0$ gives the continuity equation

$$\frac{\partial \rho}{\partial t} + \frac{\partial \rho U_1}{\partial x_1} + \frac{\partial \rho U_2}{\partial x_2} + \frac{\partial \rho U_3}{\partial x_3} = 0 \quad \text{or} \quad \frac{\partial \rho}{\partial t} = -(\nabla \cdot \rho U) \quad (2.3)$$

In CFD programs, the continuity equation is often combined with the momentum equations to form the Poisson equation for pressure. For constant density and viscosity, the new equation will be

$$\frac{\partial}{\partial x_i} \left(\frac{\partial P}{\partial x_i} \right) = - \frac{\partial^2}{\partial x_i \partial x_j} (\rho U_i U_j) \quad (2.4)$$

This equation has more suitable numerical properties and can be solved by proper iteration methods. From the above equation, the pressure P can be derived.

- **Navier-Stokes equations**

The momentum balance for the U_i momentum over the volume $\Delta x_1 \Delta x_2 \Delta x_3$ is given by

$$\left\{ \begin{array}{l} \text{rate of} \\ \text{momentum} \\ \text{accumulation} \end{array} \right\} = \left\{ \begin{array}{l} \text{rate of} \\ \text{momentum} \\ \text{in} \end{array} \right\} - \left\{ \begin{array}{l} \text{rate of} \\ \text{momentum} \\ \text{out} \end{array} \right\} + \left\{ \begin{array}{l} \text{sum of forces} \\ \text{acting on} \\ \text{the system} \end{array} \right\} \quad (2.5)$$

Newton's law requires that the change in momentum in each direction should be balanced by the forces acting in that direction. A balance for the momentum of the velocity component in the x_1 direction, i.e. U_1 , is written

$$\begin{aligned} \Delta x_1 \Delta x_2 \Delta x_3 \frac{\partial \rho U_1}{\partial x_1} = & \Delta x_2 \Delta x_3 [(\rho U_1 U_1)|_{x_1} - (\rho U_1 U_1)|_{x_1 + \Delta x_1}] - \tau_{11}|_{x_1} + \tau_{11}|_{x_1 + \Delta x_1} \\ & \Delta x_1 \Delta x_3 [(\rho U_2 U_1)|_{x_2} - (\rho U_2 U_1)|_{x_2 + \Delta x_2}] - \tau_{21}|_{x_2} + \tau_{21}|_{x_2 + \Delta x_2} \\ & \Delta x_1 \Delta x_2 [(\rho U_3 U_1)|_{x_3} - (\rho U_3 U_1)|_{x_3 + \Delta x_3}] - \tau_{31}|_{x_3} + \tau_{31}|_{x_3 + \Delta x_3} \\ & \Delta x_2 \Delta x_3 [(P)|_{x_1} - (P)|_{x_1 + \Delta x_1}] + \Delta x_1 \Delta x_2 \Delta x_3 \rho g_1 \end{aligned} \quad (2.6)$$

This formulation corresponds to a finite-volume formulation of the momentum equations for a hexahedral mesh aligned with the coordinate axis. Dividing by $\Delta x_1 \Delta x_2 \Delta x_3$ and taking the limit $\Delta x \rightarrow 0$ gives, for all three components, the Navier-Stokes equations, that in tensor notation read as

$$\frac{\partial U_i}{\partial t} + U_j \frac{\partial U_i}{\partial x_j} = -\frac{1}{\rho} \frac{\partial P}{\partial x_i} + \frac{1}{\rho} \frac{\partial \tau_{ij}}{\partial x_j} + g_i \quad (2.7)$$

From the above equation, the three components of the velocity U_1 , U_2 and U_3 can be derived.

- **Mixing**

In this scenario, since the mixing happens on a macro scale, the β -PDF is not necessary, so the model focuses exclusively on solving the concentration equations.

- **Reactions**

Since the reaction happens inside the catalyst, the cells are too small to be solved by a CFD program, so the reaction is computed through a balance between mass transfer and reaction rate.

2.2 Reactor dimensions

The reactor dimensions were computed starting from the Poiseuille equation (Eq. 2.8), imposing the pressure drop to be $\Delta P = 600 \text{ Pa}$, and solving for the channel flow speed.

$$v_{ch} = \frac{\Delta P d^2}{32 \mu (T_{in}) Z_{tot}} = 5.2722 \frac{m}{s} \quad (2.8)$$

Where $d = 0,005 \text{ m}$ is channel dimension, $Z_{tot} = 2 \cdot Z_{cat} = 2 \cdot 0.75 \text{ m}$ is catalyst total height, μ is the flow viscosity evaluated at the inlet temperature.

Subsequently the inlet flow rate is computed thanks to the ideal gas law (Eq. 2.9), using the values in (Tab. 1) and $T_{std} = 293 \text{ K}$, $P_{std} = 1 \cdot 10^5 \text{ Pa}$.

$$Q_{in} = \frac{Q_{std} T_{in} P_{in}}{T_{std} P_{std}} = 406.601 \frac{m^3}{s} \quad (2.9)$$

Now is finally possible to compute the number of modules that must be used in the catalyst, by dividing the required area of the catalyst, to have $\Delta P = 600 \text{ Pa}$, with the area of the single module.

$$N_{mod} = \frac{\frac{Q_{in}}{v_{ch}}}{d^2 N_{chan} N_{elem}} = 68.55 \quad (2.10)$$

Now, rounding the value found in (Eq. 2.10) to 70, knowing the modules dimensions, the catalyst dimensions can be computed:

$$L = 13.3 \text{ m} \quad H = 9.6 \text{ m}$$

2.3 Pressure drop

The pressure drop induced by the catalysts could be simulated in Fluent by defining the catalytic regions as porous regions and adding a momentum source term. In this way three new characteristics of the catalyst needed to be defined: the viscous loss term, the inertial loss term and the porosity. In this simulation, since the catalyst consists in a multitude of small channels perpendicular to the flow direction, and at the same time the flow is slow in these channels, the viscous losses will be way higher than the inertial losses, thus the inertial term can be discarded. The value of D was computed through the Poiseuille and Darcy equations in (Eq. 2.11), where $A_{tot} = Q_{in}/v_{ch}$. Subsequently the porosity was also computed as a void fraction (Eq. 2.12).

$$D = \frac{1}{\alpha} = \frac{A_{tot} \Delta P}{Q_{in} \mu(T_{in}) Z_{tot}} = 1.28 \cdot 10^6 \frac{1}{m^2} \quad (2.11)$$

$$\varepsilon = \frac{V_{void}}{V_{mod}} = \frac{V_{ch} N_{ch} N_{elem}}{V_{elem} N_{elem}} = \frac{(d^2 Z_{cat}) N_{ch} N_{elem}}{(b_{elem} h_{elem} Z_{cat}) N_{elem}} = 0.6944 \quad (2.12)$$

Since the gas flows mainly in z-direction, and the channels do not allow the gas to flow in other directions, then the losses are not equal in every direction, but they are mainly present in the z-direction. Moreover, permeability is high along the channel and very low cross the channel. In order to consider this effect in fluent the viscous resistance was set as a very big value in the perpendicular directions, while in z-direction was set at the value computed in (Eq. 2.11), while the inertial resistance was set to 0 in every direction. Subsequently the value of D was tuned inside Fluent in order to fulfill the constrain on the pressure drop. The tuning was carried out by running simulations at first with air, and then with the flue gasses mixture, changing D each simulation, to reach the desired result. The final value, computed using the NO_x and NH_3 mixture, was:

$$D_{final} = 1.44 \cdot 10^6 \frac{1}{m^2}$$

2.4 Flow regimes

To define the flow regime inside the reactor, the Reynolds number was computed for every different zones of the reactor.

To compute the Reynolds number is necessary to first compute the density of the flow, which was approximated to the density of the bulk flow N_2 , using the perfect gas law, as can be seen in (Eq. 2.13).¹

$$\rho_{N_2} = \frac{P_{in} MM_{N_2}}{RT_{in}} = 0.5197 \frac{kg}{m^3} \quad (2.13)$$

Then, subsequently, the Reynolds number for the inlet and for the single catalyst channel can be computed as shown in (Eq. 2.14). As it can be seen, the flow is clearly turbulent at the inlet, being higher than the transition value $Re_{in} \gg 4000$, while in the channels the flow is clearly laminar being lower than the transition value $Re_{ch} \ll 2000$.

$$Re_{in} = \frac{\rho_{N_2} Q_{in} D_{car}}{A_{in} \mu(T_{in})} = 5.9414 \cdot 10^5 \quad \text{and} \quad Re_{ch} = \frac{\rho_{N_2} v_{ch} D_{car}}{\mu(T_{in})} = 231.1175 \quad (2.14)$$

Where $D_{car} = 6m$ and $A_{in} = d^2 = 36m^2$ and the other quantities were already computed in (Eq. 2.8,2.9).

¹ $MM_{N_2} = 28.02 \cdot 10^{-3} \frac{kg}{mol}$ is the molar mass of N_2

3 Simulation

3.1 Model formulation and assumptions

In order to solve the equations stated in (Par. 2.1), different model were used together with some simplifying assumptions.

3.1.1 Turbulence modelling

The chosen turbulence model is the realizable $k - \varepsilon$ model, a modification of the standard $k - \varepsilon$ model. This model is preferred for its ability to provide improved simulations for swirling flows and flow separation, making it a reliable and commonly used option among available turbulence models.

The RNG $k - \varepsilon$ model was not chosen due to its stability problems, RSM model was not chosen because it was too computationally expensive, standard $k - \varepsilon$ model was not chosen since the flow presented re-circulation. $k - \omega$ model was used in a few tests to see its behaviour with the separated flows. LES and DNS models were discarded due to the complexity of the geometry.

3.1.2 Discretization scheme

Regarding the discretization schemes, initially, the model was run using the first-order upwind scheme. This scheme aims to ensure physical reliability for convective flows by setting the face value between two cells equal to the value of the nearest upstream cell, resulting in a bounded and transportive scheme. Recognizing that the first-order upwind scheme tends to overestimate the transport of entities in the flow direction, leading to numerical diffusion, starting from the results of the first-order, a transition was made to the second-order upwind scheme. This approach predicts face values using information from two upwind cells, aiming to address the issue of numerical diffusion of the first-order scheme, but resulting in an unbounded scheme, meaning that the model can result unstable and needs more computational time.

3.1.3 Pressure-velocity coupling

To solve the pressure-velocity coupling, the SIMPLE method was chosen. Even if this method neglects velocity corrections and may need under-relaxation, it is one of the most commonly used methods and works well for simple geometries.

3.1.4 Mixing

The presence of the catalyst shows that the reaction (Eq. 1.1) could not take place inside the limited reactor domain, and thus needs to be sped up. As a consequence the reaction can be regarded as a slow reaction, so the mixing happens at a macroscopic level, this means that it can be modelled without the need of creating a β -PDF.

3.1.5 Reaction

Since the walls inside the catalyst are too small to be solved with CFD, an assumption is made, saying that in steady state, for each computational cell, the reaction in the wall and the mass transfer through the wall are in balance, in this way the reaction can be simulated with the given formulas.

3.2 Initial and boundary conditions

3.2.1 Chemical quantities

The initial conditions for the chemical quantities to insert in the domain were chosen from (Tab. 1), imposing the respective mole fraction as inlet initial condition, for each production case. When simulating perfect mixing, the NH_3 mole fraction was assumed to be equal to the NO one, since, in the chemical reaction of the catalyst (Eq. 1.1), they had the same stoichiometric coefficient. Instead, when using the injectors, the quantity of NH_3 was put to 0 in the overall inlet, because it was inserted in the domain through the injectors. The flowrate of NH_3 passing through the injectors is computed in (Par. 3.2.2).

3.2.2 Injectors initial conditions

The ammonia injectors, to work properly, needed initial conditions of their own. The turbulence conditions were imposed equal to the general inlet conditions, but a new source term for NH_3 was added. In each injector the NH_3 volumetric mass flowrate was specified as a constant value. As a first guess try (Eq. 3.1) was used, where, to compute the NH_3 flowrate, it was assumed that the molar rate of NO was equal to the molar rate of NH_3 . At the same time the mass fraction of NO was taken from Fluent and it was equal to $\chi_{NO} = 5.2495 \cdot 10^{-4}$. The value was then tuned in order to respect the limit on ammonia slip and, at the same time, optimize the NO_x conversion.

$$\dot{m}_{N_2} = Q_{in} \rho_{N_2} \longrightarrow \dot{m}_{NO} = \chi_{NO} \dot{m}_{N_2} \longrightarrow \dot{n}_{NO} = \frac{\dot{m}_{NO}}{MM_{NO}} \longrightarrow \dot{m}_{inj} = \frac{\dot{n}_{NO} MM_{NH_3}}{20 V_{inj}} \quad (3.1)$$

The NH_3 flowrate for the different production cases are summed up in (Tab. 2). This flowrate was computed making the assumption that the volume of the injectors V_{inj} was the average of the volumes of the real injectors. The injectors were positioned, as shown in (Fig. 2), to have the ammonia spread as evenly as possible inside the domain.

	NH_3 Injection
100%	$0.5118 \text{ kg/m}^3\text{s}$
75%	$0.3663 \text{ kg/m}^3\text{s}$
50%	$0.2333 \text{ kg/m}^3\text{s}$
25%	$0.1228 \text{ kg/m}^3\text{s}$

Table 2: NH_3 first try injection

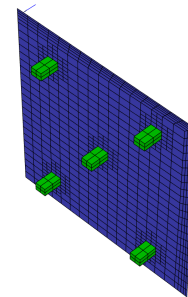


Figure 2: Injectors position

3.2.3 Inlet and outlet type

Following the data present in (Tab. 1), the absolute pressure was set as operating pressure inside Fluent, while the temperature it was imposed as inlet boundary condition. Concerning the type of boundary conditions, the type of inlet chosen was velocity inlet. Within this boundary condition, the normal velocity to the surface, turbulent viscosity, and hydraulic diameter were defined. For the outlet, the type of boundary condition chosen is pressure outlet. The two internal walls were modeled as symmetric walls with a planar symmetry, in this way only one quarter of the reactor could be studied.

3.2.4 Turbulence boundary conditions

In according with (Par. 3.2.3), the related quantities are calculated as follows:

- The normal inlet velocity of the flow was set in accordance to (Eq. 3.2), using the inlet flowrate and the inlet area.
- Turbulent intensity was computed through the correlation shown in (Eq. 3.2)
- The hydraulic diameter was set equal to the inlet edge, meaning $D_{hyd} = 6m$.

In (Eq. 3.2) are shown the full production case's results, when studying the other production cases all the variables were computed again.

$$v_{in} = \frac{Q_{in}}{A_{in}} = 11.3 \frac{m}{s} \quad I = 0.16 Re_{in}^{-\frac{1}{8}} \cdot 100 = 3.0366 \quad (3.2)$$

3.2.5 Wall functions

Since the $k - \varepsilon$ model requires wall functions and, given the non-parallel flow behavior near some of the walls, non-equilibrium wall functions were utilized instead of standard wall functions. Furthermore, it is essential to monitor the y^+ value and compare it with specific limits. Commonly, y^+ should be between $20 - 30 < y^+ < 100$.

3.3 Sanity check

3.3.1 Mesh quality

In order to have reasonable results and fast computational times, the mesh quality must be checked. The mesh was designed using hexaedral elements, which is the most flexible and reliable element type. Analysing the adapted mesh quality, skewness is found to be $Skewness^{max} = 0.6012 < 0.95$ and $Skewness^{avg} = 0.21877 < 0.33$, which denotes a good mesh. Instead the maximum aspect ratio is found to be $Aspect Ratio = 34.11 > 10$, but the majority of the cells is well below the limit, indeed 84% of the cells have an aspect ratio smaller then 10, as can be seen in (Fig. 3). Moreover the cells that present a high aspect ratio are the cells in the inlet/outlet channel, where the flow is mainly aligned to the mesh, meaning that these cells with high aspect ratio are not problematic.

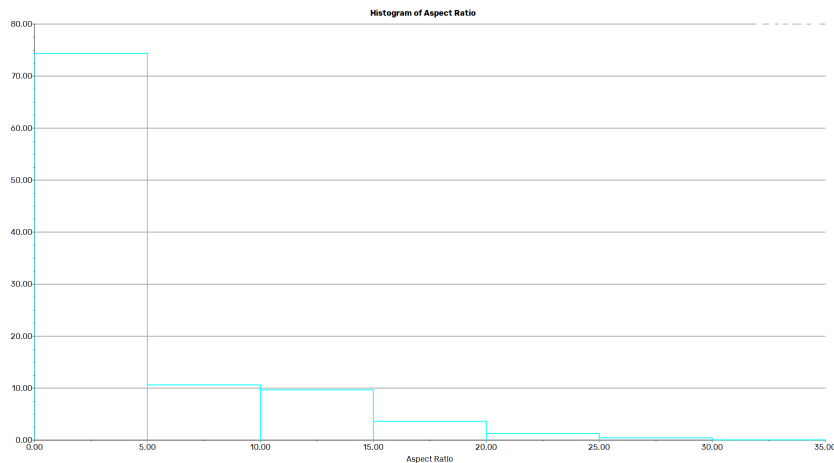


Figure 3: Aspect ratio histogram

3.3.2 Mesh independence

Since the injectors were implemented in the model by selecting a single cell per injectors, and then activating a source term for NH_3 , it was not possible to mesh the model with different element dimensions and thus investigating mesh independence in this way. This is due to the fact that the injectors would not be in the same position in every mesh, moreover they would have different volumes, meaning different NH_3 injection in each cell.

It was decided to study mesh independence by refining the mesh in different ways. The refinement was done by adapting the cells that had a particular NH_3 mole fraction during full production with NH_3 injection of $0.51 \text{ kg/m}^3\text{s}$, using the command "more than" selecting a different number of cells each time. In (Tab. 3) the results for the different meshes are presented. It can be seen that, for refined meshes, the results do not depend by the number of cells refined. This is shown by the fact that all the values studied are very close independently of the mesh, in particular the ammonia slip values. For later calculations it was chosen to use the mesh where 5592 cells were split since it had a faster computational time respect to the finer mesh, in both first and second order discretization. At the same time, suffered less of numerical diffusion, when using the first order, respect to the coarser mesh.

	Cells	Diff	Criterion	NO_x Reduction	NH_3 Slip	ΔP Drop
No ref.	46729	//	//	473.8288 ppm	14.4404 ppm	592.361 Pa
1606 cells	57971	+11242	0.00082	472.2039 ppm	14.8275 ppm	592.353 Pa
5592 cells	85873	+39144	0.00050	471.0987 ppm	14.9687 ppm	593.550 Pa
9000 cells	109729	+63000	0.00039	471.0907 ppm	14.9888 ppm	593.917 Pa

Table 3: Mesh independence study

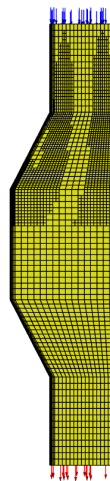


Figure 4: Refined mesh used for calculations

3.3.3 Turbulent boundary conditions influence

One important step of evaluating if the model is correct is to analyse the influence of the inlet boundary conditions for turbulence. This is done by computing the length of the large eddies, as shown in (Eq. 3.3). The length of the large eddies entering the system, which is 2.55 m , indicates that these eddies influence the flow up to approximately half of the inlet length. Beyond this point, the flow dynamics are resolved by the software simulation, meaning that the domain is not significantly impacted by the turbulent conditions at the inlet boundary.

$$l = \frac{k^{3/2}}{\varepsilon} = 2.55\text{ m} \quad (3.3)$$

3.3.4 Values check

In order to understand if the model is fit to simulate the problem, a sensible analysis on the results must be carried on. For example the velocity changes in a reasonable way, decelerating to values close to the ones hand calculated inside the catalyst. In the full production case the velocity in the catalyst calculated by hand was equal to $v_{ch} = 5.2722\text{ m/s}$ (Eq. 2.8), while the velocity found from Fluent is around 5 m/s , as shown for example in (Fig. 7). At the same time, after tuning the reactor to have a $\Delta P \leq 600\text{ Pa}$, it can be seen that the pressure drop changes in a reasonable way when changing the production cases, decreasing by decreasing the inlet flowrate, as can be seen in (Tab. 8). NO_x conversion happens in a reasonable way, allowing for a small ammonia slip, as can be expected, since the reaction is not perfect.

3.3.5 Material balance

Since the main aim of this model is to compute how much NH_3 is needed to reduce NO_x emissions, the modeling of the reaction plays a crucial role in ensuring the successful execution of the simulation. To check if the reaction is being modelled correctly is important to check the material balance at the inlet and at the outlet of the domain. To accomplish this, is needed to compute the mole fraction difference of NH_3 and NO between a region where they are already enough mixed, and the outlet. As region where NH_3 and NO already mixed it was chosen a plane distant 3 m in z direction from the inlet, parallel to it. Since the mole fraction difference shows how much of a specie has reacted, and basing the analysis on (Eq. 1.1), it is clear that the mole mole fraction difference of NH_3 and NO must be equal, since they have the same stoichiometric coefficient. Material balance was reached since, running the model with different initial conditions on NH_3 inlet and in different production cases, the mole fraction difference of NH_3 and NO was always very close, as shown in (Tab. 4). In order to check if the solution converged, NH_3 and NO values were monitored at the outlet during each iteration and the calculations were stopped only when the monitored quantities reached a plateau and their value was constant for a certain number of iterations.

	NH_3 Inj	NO_x In	NO_x Out	NO_x Diff	NH_3 In	NH_3 Out	NH_3 Diff
100%	0.5106	$4.899 \cdot 10^{-4}$	$1.850 \cdot 10^{-5}$	$4.714 \cdot 10^{-4}$	$4.912 \cdot 10^{-4}$	$1.522 \cdot 10^{-5}$	$4.760 \cdot 10^{-4}$
100%	0.5100	$4.899 \cdot 10^{-4}$	$1.881 \cdot 10^{-5}$	$4.711 \cdot 10^{-4}$	$4.906 \cdot 10^{-4}$	$1.497 \cdot 10^{-5}$	$4.756 \cdot 10^{-4}$
75%	0.3570	$4.799 \cdot 10^{-4}$	$2.455 \cdot 10^{-5}$	$4.554 \cdot 10^{-4}$	$4.702 \cdot 10^{-4}$	$1.043 \cdot 10^{-5}$	$4.597 \cdot 10^{-4}$
75%	0.3655	$4.799 \cdot 10^{-4}$	$1.800 \cdot 10^{-5}$	$4.619 \cdot 10^{-4}$	$4.814 \cdot 10^{-4}$	$1.490 \cdot 10^{-5}$	$4.665 \cdot 10^{-4}$
50%	0.2327	$4.199 \cdot 10^{-4}$	$1.612 \cdot 10^{-5}$	$4.038 \cdot 10^{-4}$	$4.210 \cdot 10^{-4}$	$1.284 \cdot 10^{-5}$	$4.081 \cdot 10^{-4}$
50%	0.2351	$4.199 \cdot 10^{-4}$	$1.339 \cdot 10^{-5}$	$4.065 \cdot 10^{-4}$	$4.253 \cdot 10^{-4}$	$1.488 \cdot 10^{-5}$	$4.104 \cdot 10^{-4}$

Table 4: Material balance

4 Tabular results

4.1 Reactor dimensions

The reactor dimensions are presented in (Tab. 5, 6).

Regarding the modules arrangement it was chosen this format in order to have the reactor as more close to a square as possible, in order to maximise its efficiency.

N modules	Arrangement	Section
70	7x10	9.6x13.3m

Table 5: Reactor sectional arrangement

Since the inlet and outlet lengths had to be at least 4m long, it was chosen to design them at the lowest length possible in order to have more space for the rest of the reactor. The void between the diffuser and the first catalyst was designed bigger than the others in order to have the flow as straight as possible when entering the catalysts, since they were placed out of the re-circulation zone, as can be seen in (Fig. 8). The names of the volumes are found in (Fig. 5).

Inlet/Outlet	Diffuser/Collector	Catalyst	Void 1	Void 2	Void 3	Total
4m	3.5m	1.65m	1.05m	0.75m	0.5m	20m

Table 6: Reactor height

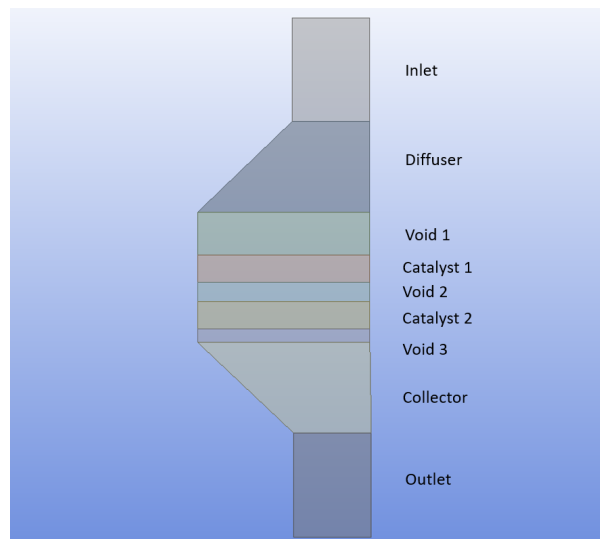


Figure 5: Zones names

4.2 Temperature influence

To investigate how the inlet temperature influences the reactor, tests with different inlet temperature, assuming perfect mixing, were carried out. The results are shown in (Tab. 7).

	Temperature	NO_x Reduction	NH_3 Slip	ΔP Drop
Perfect mixing	678.15 K	489.7486 ppm	0.2522 ppm	612.134 Pa
Perfect mixing	658.15 K	489.7944 ppm	0.2065 ppm	602.117 Pa
Perfect mixing	638.15 K	489.8363 ppm	0.1664 ppm	592.564 Pa
Perfect mixing	618.15 K	489.8666 ppm	0.1359 ppm	582.362 Pa
Perfect mixing	598.15 K	489.8841 ppm	0.1183 ppm	572.061 Pa

Table 7: Inlet temperature influence

4.3 Production cases

In (Tab. 8) the results for the different production cases are presented.

	NH_3 Injection	NO_x Reduction	NH_3 Slip	ΔP Drop
Perfect mixing	0.00049 mol/mol	489.8363 ppm	0.1665 ppm	592.564 Pa
Injectors 100%	0.5100 kg/m ³ s	471.0987 ppm	14.9687 ppm	593.551 Pa
Injectors 75%	0.3655 kg/m ³ s	461.9127 ppm	14.9023 ppm	400.552 Pa
Injectors 50%	0.2351 kg/m ³ s	406.5409 ppm	14.8834 ppm	266.489 Pa
Injectors 25%	0.1247 kg/m ³ s	369.4094 ppm	14.1402 ppm	144.842 Pa

Table 8: Results for different production cases

In this analysis, efforts were made to stay as close as possible to the 15 ppm limit for NH_3 slip. Another potential solution could have been to establish a safety margin within our system, aiming for a target of around 14 ppm, to prevent instances where ammonia slip exceeds the allowable limit during possible fluctuations of NO_x happening during production.

5 CFD-results

5.1 Full production

In (Fig. 6) the absolute pressure is presented. It is clear that the pressure from the inlet remains uniform up until the first catalyst, where a first pressure drop takes place. Then, when the flow encounters the second catalyst, a second, smaller, pressure drop takes place. After it the pressure remains uniform up until the outlet.



Figure 6: Absolute pressure contour

In (Fig. 7) the velocity vectors are shown for the full production case. It is clear that the flow slows down while passing through the diffuser and subsequently its velocity is low when entering the catalyst, confirming the calculations made in (Eq. 2.14), which stated the presence of laminar flow inside the catalyst. At the same time the flow accelerates while passing through the collector, and is fastest when entering the outlet channel.

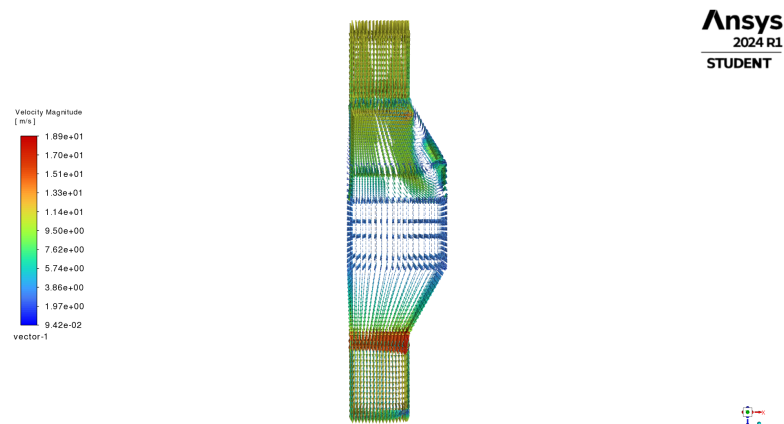


Figure 7: Velocity vectors

Some re-circulation is present in the diffuser and in the outlet channel, this is visible in (Fig. 8). Re-circulation origins due to the reactor's sharp edges and rough changes of geometry. The design was not optimized, to have better flow conditions it is advised to revisit the reactor shape and design.

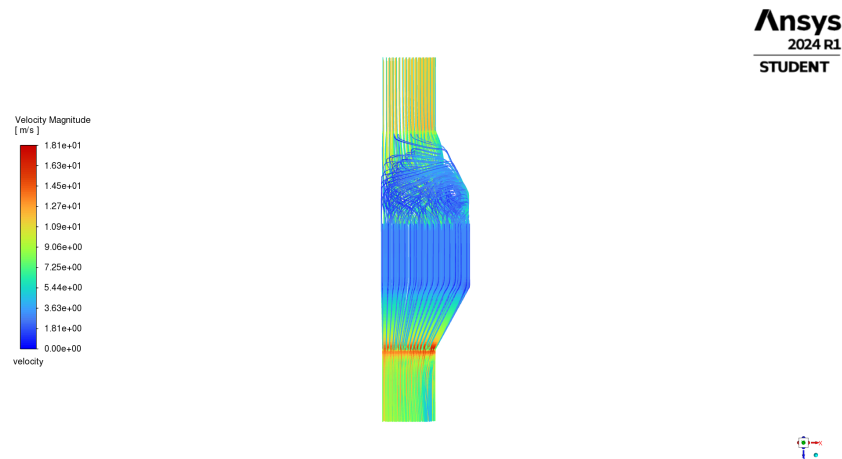


Figure 8: Pathlines colored by velocity

5.2 Reactions

In (Fig. 9) the mole fraction of NO across the domain is presented. It is interesting to notice how the majority of NO reduction happens at the first catalyst, and then is further reduced by second catalyst, with a very small quantity of NO actually exiting the reactor. This shows the importance of the catalyst presence, without which it would not be possible to have a so fast reaction, helping to reduce NO .

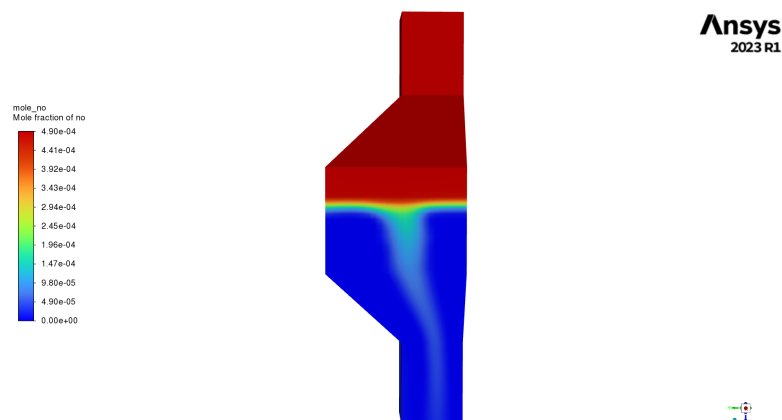


Figure 9: Mole fraction of NO contour

The reaction rate of NH_3 is shown in (Fig. 10). The majority of the reaction happens at the first catalyst. The red zones, indicate where the reaction rate is higher, this shows the importance of the injector's positions and of their initial conditions.

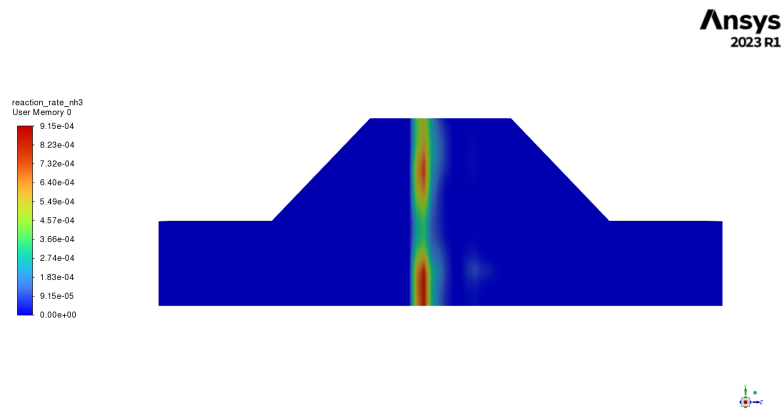


Figure 10: Reaction rate NH_3 contour

6 Discussion

6.1 Mesh independence

Since the main aim of the SCR is to convert NO_x by using NH_3 , it was chosen as mesh refinement criterion the value of NH_3 mole fraction, in order to simulate as better as possible the mixing before the catalyst. The procedure of refinement is described in in (Par. 3.3.2).

Small value changes in the results were of course expected, since a finer mesh meant that mixing was simulated better, but the results changes are minimal. Since $1\text{ ppm} = 1\text{ mole fraction} \cdot 10^6$, this means that the results in fluent change in the order of 10^{-6} or 10^{-7} , which is very small, meaning that mesh independence is reached.

Other criteria to refine the mesh could have been used, such as velocity gradients or turbulent energy gradients, they would help, but probably they would not be as useful as the criterion used, since the aim of the model was to simulate the reactions inside the SCR and not the kinetics and the flow conditions inside it.

6.2 Wall functions

Since the conditions of parallel flow to the walls was not fulfilled, standard wall function could not be used, so non-equilibrium wall functions needed to be used, as discussed in (Par. 3.2.5). As can be seen in (Fig. 11), the majority of wall cells were indeed in the $30 < y^+ < 100$ range, or very close to it, so this means that the mesh was appropriate enough to be used with non-equilibrium wall functions.

Moreover since separation occurred inside the domain, both at the diffuser and at the the outlet channel, standard wall functions were not recommended.

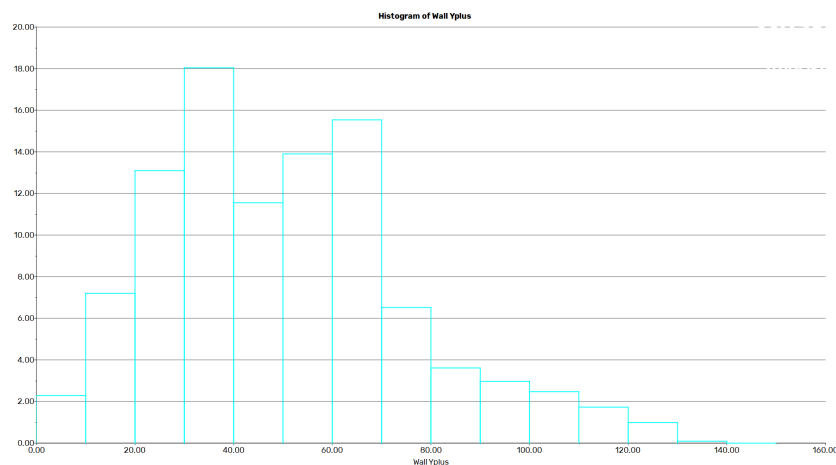


Figure 11: y^+ values

6.3 Inlet conditions

To check how the choice of inlet conditions influenced the reactor performances, a plane was created 3m from the inlet and the velocity profile was studied, in particular v_x , v_y , v_z , k and ε were sampled, these variables were used as the new inlet conditions for the new simulation. In this way the influence of the turbulent inlet boundary conditions are mitigated, since the values used are not anymore an assumption computed by hand but they are actual number computed where the flow is already well developed. The results were affected by the choice of inlet conditions as can be seen in (Tab. 9). Even if the turbulent initial conditions do not influence the flow too deep into the system, as discussed in (Par. 3.3.3), it is still enough to affect the reactor performance. In particular, these new profile conditions tend to limit the conversion, allowing for a higher ammonia slip, but do not affect much the pressure drop.

	NH_3 Inj	NO_x Reduction	NH_3 Slip	ΔP Drop
Standard inlet	$0.51 \text{ kg/m}^3\text{s}$	471.0987 ppm	14.9687 ppm	593.55 Pa
Profile inlet	$0.51 \text{ kg/m}^3\text{s}$	463.2627 ppm	19.7552 ppm	593.86 Pa

Table 9: Comparison between inlet boundary conditions

6.4 Turbulence model and discretization scheme independence

In order to define turbulence model independence, the $k - \omega$ model, with SST wall functions was tested. A new mesh was created, with a different inflation, in order to fulfill the model's requirements of $y^+ < 5$, while all the other parameters were set as in the mesh used in the calculations with the realizable $k - \varepsilon$. The same inlet boundary conditions were imposed into the $k - \omega$ model, in this way a fair comparison could be made. Unfortunately the model converged only when using first order methods, but numerical diffusion was present, when using second order the solution could not converge with this mesh since some reversed flow occurred, probably due to the presence of the big re-circulation zone. Thus the independence from the turbulent method could not be assessed.

Regarding the dependence on the discretization scheme, using the first-order upwind scheme, the simulations converged quickly, but the results had a lot of numerical diffusion. So starting from the results of the first-order, a transition was made to the second-order upwind scheme, thus leading into slower simulations, but without diffusion.

A test was carried out, in the case of full production, using the QUICK and MUSCL scheme, the results are presented in (Tab. 10). The schemes proved to be slower, needed more iterations to reach a stable solution and needed lower under-relaxation factors, moreover the results were slightly affected by these choices of schemes. Thus, there emerges a slight dependence on the choice of discretization scheme. Based on this, the rest of the simulations were carried out using the second order upwind scheme, since it is the faster and more reliable scheme.

	Scheme	NH_3 Inj	NO_x Reduction	NH_3 Slip	ΔP Drop
Full production	2° Upwind	$0.51 \text{ kg/m}^3\text{s}$	471.0987 ppm	14.9687 ppm	593.55 Pa
Full production	QUICK	$0.51 \text{ kg/m}^3\text{s}$	473.5066 ppm	14.2512 ppm	596.11 Pa
Full production	MUSCL	$0.51 \text{ kg/m}^3\text{s}$	472.5376 ppm	15.6198 ppm	591.53 Pa

Table 10: Different discretization schemes

6.5 Temperature and mixing influence

The influence of inlet temperature was investigated in (Par. 4.2). By increasing the temperature, the conversion proved to be slightly worse, since the value of NO_x converted got lower and the NH_3 slip higher, and vice versa. Nevertheless the values are very close to each other, meaning that temperature does not influence much the conversion. This is due to the assumption of perfect mixing: if the test was carried out using the injectors the influence of temperature would probably be higher.

Regarding the pressure drop instead, it is influenced by temperature, an increase of 20 K resulted in an increase of 10 Pa . This means that it is important to control the temperature fluctuations in real life, in order to always respect the maximum pressure drop imposed to the SCR.

The primary influence of mixing is attributed to the presence of injections instead of perfect mixing, leading to a non-uniform distribution of NH_3 along the SCR bed. In the absence of perfect mixing, the risk of ammonia slip increases. When ammonia is not uniformly distributed, certain zones may have lower-than-expected NO_x levels, leading to an excess of unreacted ammonia and resulting in ammonia slip at the outlet of the SCR, resulting in a not perfect NO_x conversion.

When instead perfect mixing is simulated, all the ammonia was already mixed with the bulk flow, meaning that it could almost entirely react inside the catalysts, thus the value of ammonia slip was very low compared to the case with injectors.

6.6 Conversion limitation

The conversion is mainly limited by mixing, this can be seen in the results difference when modelling the reactor with perfect mixing and when instead injector are used and thus mixing is modelled. The differences are shown in (Tab. 8), regarding the pressure drop the difference is minimal, but instead when looking at the NO_x reduction and NH_3 slip values the change is important. When using injectors, all the species directly mix inside the reactor, the mixing will of course not be perfect, meaning that not all the ammonia will mix with the bulk flow. The part of ammonia that will not be mixed with NO_x will not react with it when passing through the catalyst, meaning that some of the ammonia will also exit the reactor, as ammonia slip. When simulating with perfect mixing instead, almost all the ammonia will react in the catalysts, so ammonia slip will be very small, as can be seen in (Tab. 8), where the quantity of NH_3 slip is almost 3 orders of magnitudes smaller than the model using injectors. The influence of mixing can also be seen in (Tab. 3), where by refining the mesh, the more NH_3 was mixed with NO_x , the more the model would be closer to reality. The more precisely the mixing was simulated, the further it got from perfect mixing, meaning that NH_3 slip gets higher and NO_x reduction gets lower. Kinetics will of course influence the conversion, because if, for example, the flow is very slow, then mixing and reaction will happen slower, but kinetics influence will be smaller than the mixing. Since the aim of the model is to reproduce the results of a chemical reaction happening inside the reactor, then mixing will be more important and more limiting in the outcomes of the simulation.

6.7 2D mesh prediction

The SCR cannot be accurately predict with a 2D mesh for several reasons:

- **Geometry and flow behaviour**

The SCR reactor geometry influences greatly the model: injection nozzles, catalyst layers, diffuser and collector significantly affect the performance of the reactor because of their great influence the flow's behaviour. The spatial distribution of the catalyst channel changes how the catalyst interacts with the flow. This interaction depends on the spatial arrangement and is not possible to simulate in 2D. Moreover, the positioning of the injectors directly influence how NH_3 mixes with the bulk flow, thus mixing cannot be solved by a 2D mesh. Lastly, the flow inside the reactor, with the current geometry design, cannot be predicted with a 2D mesh since it is highly three-dimensional: as can be seen in (Fig. 8), separation and a big re-circulation zone are present, and this flow behaviour can be described well only using a 3D mesh.

- **Catalyst Interaction**

SCR involves surface reactions occurring on the catalyst. These reactions depend as much on the catalyst composition as on local flow conditions, including velocity profile, turbulence, and species concentration on the catalyst surface. All these variables are oversimplified in a 2D mesh simulations.

Hence, a three-dimensional mesh of the catalyst bed is necessary in order to understand the reaction kinetics, the interaction between the gasses and to accurately predict the reactor efficiency and behaviour.

7 Conclusions

This study addressed the design and CFD simulation of a Selective Catalytic Reduction (SCR) system, whose aim consists in mitigating NO_x emissions from power plant flue gasses, by transform NO_x into N_2 and H_2O , thanks to the injection of NH_3 into the system. The study mainly focused in modelling the injection of NH_3 into the bulk flow, and the kinematic and chemical behaviour of the entire flow inside the reactor domain. The model parameters, such as NH_3 injection and permeability, were successfully tuned in order to fulfill the constraints given on maximum pressure drop and ammonia slip, during each production case.

All the guidelines to create a good and solid CFD analysis were followed as strictly as possible. Theory was investigated in order to both define the best solution method available and, at the same time, to find important parameters or first try values for inlet conditions. This resulted in using equations such as Darcy equation, Poiseuille equation, perfect gas law, flowrate definition...

The geometry was created following all the instructions and requirements on length and pressure drop. It was subsequently discretized with a mesh designed following all the criteria necessary to have a good simulation. Hexahedral elements were selected, with element size equal to 0.3 m , inflation was used to refine the mesh in the vicinity of the walls, in order to fulfill the wall functions requirements on y^+ . The mesh quality was then checked and the cell's skewness and aspect ratio values were found to be acceptable. The mesh was refined using the gradient of NH_3 mole fraction as criterion, and after this refinement the model was found to be mesh independent.

The turbulent method chosen was realizable $k - \epsilon$, which proved to be reliable and stable enough to be used in each situation and production case, with very small adjustments to under-relaxation factors and residual in order to reach stable values of NH_3 and NO_x at the outlet. The model's independence on turbulent method could not be assessed. The only model that could be used to check for turbulent method independence was the $k - \omega$ method, but due to lack of time it was not possible to reach convergence with second order upwind schemes, thus the results were not comparable. The other turbulence methods were deemed unsuitable for modeling the SCR due to either not being able to represent separated flow or because too computationally expensive. Regarding the discretization schemes, 1° and 2° order upwind, QUICK and MUSCL were used and the results compared, turning out to be slightly dependent by the choice of scheme. Nevertheless the values were still close to each other.

The effectiveness of the SCR system heavily depends on a multitude of factors including initial condition for both turbulence and ammonia injection. Even if the length scale of the large eddies, due to the inlet boundary conditions setup, does not affect the flow too much inside the inlet channel, the inlet conditions still interfere considerably the results. When using as inlet conditions the velocity profile of the already developed flow, the results changed considerably. This shows that, to have more precise results, the every simulations should be at first run with the chosen initial condition, for each production case, then the velocity profile should be sampled and the simulation run again, in this way the results should be closer to reality.

Overall the simulation proved to be stable and good enough to be used to model the SCR behaviour at different production cases and to define the injected quantity of NH_3 that allows for the maximum NO_x conversion, while respecting the imposed limit of maximum 15 ppm of ammonia slip.

Further studies to improve the model could be done on redesigning the geometry of the reactor, in order to limit the re-circulation zone. Different positioning and quantity of the NH_3 injectors could be investigated in order to optimize mixing. A more suitable mesh for $k - \omega$ method could be created, to define if turbulence method independence was reached. To have results closer to reality it is suggested to study the entire SCR system, comprehensive of the ducts and fins that bring the flow to the catalysts, in order to have a better simulation of the flow behaviour in the whole domain.

In conclusion, the successful implementation of a SCR's CFD simulation is based on a detailed understanding of chemical reaction dynamics, fluid mechanics, and numerical methods. This study aims to prove insights into the design of the reactor CFD model, to study its parameters, ensuring both environmental compliance and operational efficiency. In the future, the aim should be on refining these models to incorporate real-time operational data, machine learning techniques and dynamic adaptation to achieve improved calculations and results.

A Matlab code

In this paragraph the Matlab code used for calculations is presented.

```
clc
clear
close all

%% Data

% Fluid
P_drop = 600; % [Pa]
mu = @(T) 1.23*10^-6*T.^0.6; % [Pas]

% Standard Conditons
T_std = 293; % [K]
P_std = 1e5; % [Pa]
Q_std = [683000, 493000, 356000, 206000]/3600; % [Nm^3/s]

% Inlet
T_in = [365, 340, 310, 290] + 273.15; % [K]
P_in = [98400, 99000, 99400, 99650]; % [Pa]

% SCR
d = 5e-3; % [m] Channel dimension
N_elem = 72; % [1] Elements per module
N_chan = 625; % [1] Channel per element
Z = 2*0.75; % [m] Catalysts heighth
L_mod = 1.9; % [m]
H_mod = 0.96; % [m]

%% Task 1) - Reactor Dimensions

v_channel = P_drop*d^2/(32*mu(T_in(1))*Z); % [m/s]
Q_in = Q_std.*T_in.*P_in./(P_std*T_std); % [m^3/s]

A_tot = Q_in(1)/v_channel; % [m^2]
N_mod = A_tot/(d^2*N_chan*N_elem); % [1]

L_tot = L_mod*7/2;
H_tot = H_mod*10/2;
```

```

%% Task 2) - Pressure Drop

V_mod = 0.150*0.150*0.75*N_elem;
V_cha = d^2*0.75;

V_void = V_cha*N_chan*N_elem;
Porosity = V_void/V_mod;

Permeability = Q_in(1)*mu(T_in(1))*2*0.75/(A_tot*P_drop);
D = 1/Permeability;

%% Task 3) Reynolds

MM_N2 = 28.02e-3; % [kg/mol]
rho_N2 = P_in*MM_N2./(8.314*T_in); % [kg/m^3]

v_in = Q_in/(6*6) % [m/s]
Re_In = rho_N2*Q_in(1)*6./(6*6*mu(T_in)); % [1]

I = 0.16*Re_In.^(-1/8)*100 % [1]

Re_Ch = rho_N2*v_channel*d/mu(T_in(1)); % [1]

%% Ammonia injectors

Vol_inj = mean([0.0061577907, 0.006245183, 0.0061516819,
    0.0060520252, 0.0061394155]); % [m^3]

MM_NH3 = 0.017031; % [kg/mol]
MM_NO = 30.01e-3; % [kg/mol]
MF_NO = [0.00052495, 0.0005142397028893, 0.0004499510744928,
    0.0004070933645447];

m_n2 = Q_in.*rho_N2; % [kg/s]
m_NO = MF_NO.*m_n2;

% molar rate of NO = to molar rate of NH3
n_NO = m_NO/MM_NO;

m_amm = n_NO*MM_NH3;
m_inj = m_amm./(20*Vol_inj)

```